Convex Optimization

Homework 1

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1 Exercise 1 - Convex Sets

1. A slab, i.e., a set of the form
$$\{x \in \mathbb{R}^n \mid \alpha \leq a^T x \leq \beta\}.$$

   Answer: <u>This set is convex.</u>
   It is the intersection of two halfspaces, which are convex sets:
$$\{x \in \mathbb{R}^n \mid a^T x \leq \beta\} \cap \{x \in \mathbb{R}^n \mid \alpha \leq a^T x\}.$$
   Since the intersection of convex sets is convex, the slab is convex.

2. A rectangle, i.e., a set of the form
$$\{x \in \mathbb{R}^n \mid \alpha_i \leq x_i \leq \beta_i, ß = 1, \ldots, n\}.$$

   Answer: <u>This set is convex.</u>

$$\begin{aligned}\{x \in \mathbb{R}^n \mid \alpha_i \leq x_i \leq \beta_i,\ i = 1, \ldots, n\} &= \bigcap_{i=1}^{n} \{x \in \mathbb{R}^n \mid \alpha_i \leq x_i \leq \beta_i\} \\ &= \bigcap_{i=1}^{n} \left(\{x \in \mathbb{R}^n \mid \alpha_i \leq x_i\} \cap \{x \in \mathbb{R}^n \mid x_i \leq \beta_i\}\right).\end{aligned}$$

   It is an intersection of halfspaces, thus convex.

3. A wedge, i.e.,
$$\{x \in \mathbb{R}^n \mid a_1^T x \leq b_1,\ a_2^T x_2\}.$$

   Answer: <u>This set is convex.</u>
   It is the intersection of two halfspaces, thus convex:
$$\{x \in \mathbb{R}^n \mid a_1^T x \leq b_1\} \cap \{x \in \mathbb{R}^n \mid a_2^T x \leq b_2\}.$$

4. The set of points closer to a given point than a givenset, i.e.,
$$\{x \mid \|x - x_0\|_2 \leq \|x - y\|_2 \text{ for all } y \in S\}, \quad S \subseteq \mathbb{R}^n.$$

   Answer: <u>This set is convex.</u>

Let $y \in S$, and $x_0 \in \mathbb{R}^n$.

$$\begin{aligned}\|x - x_0\|_2 \leq \|x - y\|_2 &\Leftrightarrow (x - x_0)^T(x - x_0) \leq (x - y)^T(x - y) \\ &\Leftrightarrow x^T x - 2x_0^T x + x_0^T x_0 \leq x^T x - 2y^T x + y^T y \\ &\Leftrightarrow 2(y - x_0)^T x \leq y^T y - x_0^T x_0\end{aligned}$$

Notice that the norm inequality is equivalent to a halfspace $a^T x \leq b$, with

$$a = 2(y - x_0), \quad b = y^T y - x_0^T x_0.$$

We get the following equality:

$$\begin{aligned}\{x \mid \|x - x_0\|_2 \leq \|x - y\|_2 \text{ for all } y \in S\} &= \bigcap_{y \in S} \{x \mid \|x - x_0\|_2 \leq \|x - y\|_2\} \\ &= \bigcap_{y \in S} \{x \mid 2(y - x_0)^T x \leq y^T y - x_0^T x_0\}\end{aligned}$$

Since the set is the intersection of halfspaces, it is convex.

5. The set of points closer to one set than another, i.e.,

$$\{x \mid \operatorname{dist}(x, S) \leq \operatorname{dist}(x, T)\}, \quad S, T \subseteq \mathbb{R}^n,$$

where

$$\operatorname{dist}(x, S) = \inf\{\|x - z\|_2 \mid z \in S\}.$$

Answer: <u>This set is not convex.</u>

Let's provide an example where this set is not convex in $\mathbb{R}$.
Let

$$S = [0, 3] \cup [7, 10], \quad T = [4, 6]$$

We notice that:

$$\{x \mid \operatorname{dist}(x, S) \leq \operatorname{dist}(x, T)\} =]-\infty, 3.5] \cup [6.5, +\infty[$$

Which is clearly not a convex set.

6. The set

$$\{x \mid x + S_2 \subseteq S_1\}, \quad S_1, S_2 \subseteq \mathbb{R}^n \text{ with } S_1 \text{ convex.}$$

Answer: <u>This set is convex.</u>

$$\begin{aligned}\{x \mid x + S_2 \subseteq S_1\} &= \{x \mid x + y \in S_1, \forall y \in S_2\} \\ &= \bigcap_{y \in S_2} \{x \mid x + y \in S_1\} \\ &= \bigcap_{y \in S_2} S_1 - y\end{aligned}$$

We notice that this set is just a translation of S_1, which is a convex set. The set is an intersection of convex sets, which means that it is convex.

7. The set of points whose distance to a does not exceed a fixed fraction θ of the distance to b,i.e.,

$$\{x \mid \|x-a\|_2 \leq \theta\|x-b\|_2\}, \quad a \neq b,\ 0 \leq \theta \leq 1.$$

Answer: <u>This set is convex.</u>

First, let's consider the case $\theta = 0$. We get $\|x-a\|_2 \leq 0$. Since the norm must be non-negative, $\|x-a\|_2 = 0$, which implies $x = a$. The set $\{a\}$ contains a single point, and is thus convex.

Let's also consider the case $\theta = 1$. The inequality becomes:

$$\|x-a\|_2 \leq \|x-b\|_2$$

We immediatly notice that this is simply a special case of problem 4, where $S = \{b\}$ and $x_0 = a$. We have already proven that the set of problem 4 is convex, which means that the set for $\theta = 1$ is also convex.

We now assume $0 < \theta < 1$. Let's look at the norm inequality:

$$\begin{aligned}
\|x-a\|_2 \leq \theta\|x-b\|_2 &\Leftrightarrow \|x-a\|_2^2 \leq \theta^2\|x-b\|_2^2 \\
&\Leftrightarrow (x-a)^T(x-a) \leq \theta^2(x-b)^T(x-b) \\
&\Leftrightarrow x^Tx + a^Ta - 2a^Tx \leq \theta^2x^Tx + \theta^2b^Tb - 2\theta^2b^Tx \\
&\Leftrightarrow (1-\theta^2)x^Tx + 2(\theta^2b^T - a^T)x \leq \theta^2b^Tb - a^Ta
\end{aligned}$$

By completing the square, we get

$$\begin{aligned}
(1-\theta^2)x^Tx + 2(\theta^2b^T - a^T)x &= (1-\theta^2)x^Tx + 2\frac{\sqrt{1-\theta^2}}{\sqrt{1-\theta^2}}(\theta^2b-a)^Tx + \frac{1}{\sqrt{1-\theta^2}}(\theta^2b-a)^T(\theta^2b-a) - \frac{1}{\sqrt{1-\theta^2}}(\theta^2b \\
&= \left\|\sqrt{1-\theta^2}x + \frac{\theta^2b-a}{\sqrt{1-\theta^2}}\right\|_2 - \frac{1}{\sqrt{1-\theta^2}}(\theta^2b-a)^T(\theta^2b-a) \\
&= \sqrt{1-\theta^2}\left\|x + \frac{\theta^2b-a}{1-\theta^2}\right\|_2 - \frac{1}{\sqrt{1-\theta^2}}(\theta^2b-a)^T(\theta^2b-a)
\end{aligned}$$

Which means that:

$$\begin{aligned}
\|x-a\|_2 \leq \theta\|x-b\|_2 &\Leftrightarrow \sqrt{1-\theta^2}\left\|x + \frac{\theta^2b-a}{1-\theta^2}\right\|_2 \leq \theta^2b^Tb - a^Ta + \frac{1}{\sqrt{1-\theta^2}}(\theta^2b-a)^T(\theta^2b-a) \\
&\Leftrightarrow \left\|x - \frac{a-\theta^2b}{1-\theta^2}\right\|_2 \leq \frac{\theta^2b^Tb - a^Ta}{\sqrt{1-\theta^2}} + \frac{1}{1-\theta^2}(\theta^2b-a)^T(\theta^2b-a)
\end{aligned}$$

Notice that this is just a norm ball with

$$\text{center } x_c = \frac{a - \theta^2 b}{1 - \theta^2} \quad \text{and radius } r = \frac{\theta^2 b^T b - a^T a}{\sqrt{1 - \theta^2}} + \frac{1}{1 - \theta^2}(\theta^2 b - a)^T(\theta^2 b - a).$$

Since the norm ball set is convex, this set is also convex.

2 Exercise 2 - Pointwise Maximum and Supremum

1.
$$f(x) = \max_{i=1,\dots,k} \|A^{(i)}x - b^{(i)}\|,$$

where $A^{(i)} \in \mathbb{R}^{m \times n}$, $b^{(i)} \in \mathbb{R}^m$, and $\|\cdot\|$ is a norm on $\mathbb{R}^m$.

Answer: Let $f_i(x) = \|A^{(i)}x - b^{(i)}\|$, for $i = 1, \dots, k$. This function is the composition of the norm of an affine function, which makes it convex for every i. Since f is the maximum of the f_i convex functions, <u>f is convex</u>.

2.
$$f(x) = \sum_{i=1}^{r} |x|_{[i]} \quad \text{on } \mathbb{R}^n,$$

where $|x|$ denotes the vector with $|x|_i = |x_i|$, and $|x|_{[i]}$ is the i-th largest component of $|x|$.

Answer: We can rewrite f as:

$$\begin{aligned} f(x) &= \max\{x_{i_1} + \cdots + x_{i_r}, 1 \le i_1 \le \cdots \le i_r \le n\} \\ &= \max_{1 \le i_1 \le \cdots \le i_r \le n} \sum_{k=1}^{r} |x|_{i_k} \end{aligned}$$

f is the maximum of the sum of $\binom{n}{r}$ convex functions. Thus, <u>f is convex</u>.

3 Exercise 3 - Products and Ratios of Convex Functions

If f and g are convex, both nondecreasing (or nonincreasing), and positive functions on an interval, then fg is convex.

Answer: Let's assume that $f, g \ge 0$ on I, and $\theta \in [0, 1]$. Then, for $x, y \in I$, we denote z as:

$$z = \theta x + (1 - \theta)y$$

In order to prove that fg is convex, we want to show that:

$$f(z)g(z) \le \theta f(x)g(x) + (1 - \theta)f(y)g(y)$$

We have, by the convexity of f and g:

$$\begin{aligned} f(z)g(z) &\leq (\theta f(x) + (1-\theta)f(y))\,(g(x) + (1-\theta)g(y)) \\ &= \theta^2 f(x)g(x) + (1-\theta)^2 f(y)g(y) + \theta(1-\theta)f(x)g(y) + \theta(1-\theta)f(y)g(x) \\ &= \alpha \end{aligned}$$

We thus have:

$$\theta f(x)g(x) + (1-\theta)f(y)g(y) - \alpha = \theta(1-\theta)\,(f(x) - f(y))\,(g(x) - g(y))$$

Since we assume that both f and g are nondecreasing (or nonincreasing), notably on interval I, then we have:

$$\operatorname{sign} f(x) - f(y) = \operatorname{sign} g(x) - g(y)$$

with $\theta(1-\theta) \geq 0$, we get:

$$\begin{aligned} \theta f(x)g(x) + (1-\theta)f(y)g(y) - \alpha = \theta(1-\theta)\,(f(x) - f(y))\,(g(x) - g(y)) &\geq 0 \\ \Leftrightarrow \alpha &\leq \theta f(x)g(x) + (1-\theta)f(y)g(y) \\ \Leftrightarrow (\theta f(x) + (1-\theta)f(y))\,(\theta g(x) + (1-\theta)g(y)) &\leq \theta f(x)g(x) + (1-\theta)f(y)g(y) \\ \implies f(z)g(z) &\leq \theta f(x)g(x) + (1-\theta)f(y)g(y) \end{aligned}$$

This is the definition of convexity. We have proved that <u>fg is convex</u> if f and g are convex, both nondecreasing (or nonincreasing), and positive functions on an interval.

4 Exercise 4 - Conjugates of Functions

1. Max function:

$$f(x) = \max_{i=1,\dots,n} x_i \quad \text{on } \mathbb{R}^n.$$

Answer: The conjugate of f is

$$f^*(y) = \begin{cases} 0 & \text{if } \sum_{i=1}^n y_i = 1 \text{ and } y_i \geq 0 \text{ for all } i \\ +\infty & \text{otherwise} \end{cases}$$

Let us start with the definition of the conjugate of function $f(x)$. It is defined as:

$$f^*(y) = \sup_{x \in \mathbb{R}^n} \left\{ y^T x - f(x) \right\}$$

For the max function, this is equal to:

$$f^*(y) = \sup_{x \in \mathbb{R}^n} \left\{ \sum_{i=1}^n y_i x_i - \max_{j=1,\dots,n} x_j \right\} = \sup_{x \in \mathbb{R}^n} \phi(x, y)$$

We then determine the conditions on y in order for the supremum to be finite.
We make the asssumption that $y_k < 0$ for some index $k \in \{1, \ldots, n\}$. Let x be a vector such that, for $t > 0$:

$$x = -\delta_k t = \begin{cases} -t & \text{if } i = k \\ 0 & \text{if } i \neq k \end{cases}$$

We can then express $\phi(x, y)$ as:

$$\begin{aligned} \phi(x, y) &= \sum_{i=1}^{n} y_i x_i - \max_{j=1,\ldots,n} x_j \\ &= -y_k t - \max(0, -t) \\ &= -y_k t \end{aligned}$$

Since we assumed $y_k < 0$, then $\phi(x, y)$ is positive and approaches ∞ as $t \to \infty$. This means that if any component of y is negative, then the supremum is infinite.

Now that we established for we must have $y_i \geq 0$ for all $i = 1, \ldots, n$, let's consider another vector x. This time, we assume that all components are equal to some $t > 0$. We write x as $x = (t, t, \ldots, t)$. Our expression of $\phi(x, y)$ becomes:

$$\begin{aligned} \phi(x, y) &= \sum_{i=1}^{n} y_i t - \max_{j=1,\ldots,n} (t, t, \ldots, t) \\ &= t \sum_{i=1}^{n} y_i - t \\ &= t \left(\sum_{i=1}^{n} y_i - 1 \right) \\ &= t\alpha \end{aligned}$$

If $\alpha > 0$ and $t \to \infty$, then the supremum is infinite. If $\alpha < 0$ and $t \to \infty$, then the supremum is also infinite. Which means that α must be equal to 0, or in other words, we must have $\sum_{i=1}^{n} y_i = 1$ for $f^*(y)$ to be finite.

In conclusion, the supremum is finite if:

(a) $y_i \geq 0$ for all $i = 1, \ldots, n$
(b) $\sum_{i=1}^{n} y_i = 1$

We can now calculate the supremum under the conditions above.

Let $m = \max_{j=1,\dots,n} x_j$. We get the following inequalities:

$$\begin{aligned}
x_i \leq m &\implies y_i x_i \leq y_i m \\
&\implies \sum_{i=1}^{n} y_i x_i \leq \sum_{i=1}^{n} y_i m \\
&\implies \sum_{i=1}^{n} y_i x_i \leq m \sum_{i=1}^{n} y_i \\
&\implies \sum_{i=1}^{n} y_i x_i \leq m \\
&\implies \phi(x,y) = \sum_{i=1}^{n} y_i x_i - m \leq m - m = 0
\end{aligned}$$

For any vector x, the expression, and thus the supremum, is less than or equal to 0. We can find an x such that the expression is 0. We consider once again the vector $x = (t, t, \dots, t)$. For this x, $\phi(x,y) = t(\sum_{i=1}^{n} y_i - 1) = t(1-1) = 0$. Since the value of 0 is reached, we conclude that the supremum must be 0.

We finally get the conjugate function:

$$f^*(y) = \begin{cases} 0 & \text{if } \sum_{i=1}^{n} y_i = 1 \text{ and } y_i \geq 0 \text{ for all } i \\ +\infty & \text{otherwise} \end{cases}$$

2. Sum of largest elements:

$$f(x) = \sum_{i=1}^{r} x_{[i]} \quad \text{on } \mathbb{R}^n.$$

Answer: The conjugate of f is

$$f^*(y) = \begin{cases} 0 & \text{if } 0 \preceq y \preceq 1 \text{ and } \sum_{i=1}^{n} y_i = r \\ +\infty & \text{otherwise} \end{cases}$$

Once again, we proceed with the same steps, starting with determining the domain of the conjugate function. Let $phi(x,y) = y^T x - \sum_{i=1}^{r} x_{[i]}$. Let $x = (t, t, \dots, t)$, such that all components, and notably the r largest

components, are all equal to t. $phi(x, y)$ becomes:

$$\begin{aligned}
\phi(x, y) &= y^T x - \sum_{i=1}^{r} x_{[i]} \\
&= t \sum_{i=1}^{n} y_i - rt \\
&= t(\sum_{i=1}^{n} y_i - r) \\
&= t\alpha
\end{aligned}$$

Notice that if α is not null, then the supremum is infinite as $t \to +\infty$ or $t \to -\infty$. We must have:

$$\sum_{i=1}^{n} y_i = r$$

Now let's assume that $x = \delta_k t = (0, \ldots, t, \ldots, 0)$ for some index k and $t > 0$. We can express $\phi(x, y)$ as:

$$\phi(x, y) = ty_k - t = t(y_k - 1)$$

Notice that if $y_k > 1$ then the supremum is infinite for $t \to +\infty$. We must have $y_{[i]} \leq 1$ for all $i = 1, \ldots, r$. If we assume that $x = -\delta_k t$ for some index k and $t > 0$, then $\phi(x, y)$ becomes:

$$\phi(x, y) = -ty_k$$

If $y_k < 0$ for some $k > r$ then as $t \to +\infty$, the supremum is infinite. We conclude that y_i must be positive for all $i = r + 1, \ldots, n$. Putting this together with the previous condition, we get that $0 \leq y_i \leq 1$ must be true for all $i = 1, \ldots, n$.

In conclusion, the supremum is finite if:

(a) $\sum_{i=1}^{n} y_i = r$
(b) $0 \leq y_i \leq 1$ for all $i = 1, \ldots, n$

We can now calculate the supremum under the conditions above.
We rewrite $\phi(x, y)$ as:

$$\begin{aligned}
\phi(x, y) &= \sum_{i=1}^{n} y_{[i]} x_{[i]} - \sum_{i=1}^{r} x_{[i]} \\
&= \sum_{i=1}^{r} y_{[i]} x_{[i]} + \sum_{i=r+1}^{n} y_{[i]} x_{[i]} - \sum_{i=1}^{r} x_{[i]} \\
&= \sum_{i=1}^{r} (y_{[i]} - 1) x_{[i]} + \sum_{i=r+1}^{n} y_{[i]} x_{[i]}
\end{aligned}$$

We notice, under the conditions above, that $y_i - 1 \leq 0$, $x_{[i]}$ in the left sum term is superior to $x_{[r]}$ (under the assumption that the values are sorted from largest to smallest, without any loss of generality), $y_{[i]}$ in the right sum term is positive and $x_{[i]}$ is smaller than $x_{[r]}$ for $i = r+1, \ldots, n$. We get the following bound:

$$\begin{aligned}
\phi(x, y) &= \sum_{i=1}^{r} (y_{[i]} - 1) x_{[i]} + \sum_{i=r+1}^{n} y_{[i]} x_{[i]} \\
&\leq \sum_{i=1}^{r} (y_{[i]} - 1) x_{[r]} + \sum_{i=r+1}^{n} y_{[i]} x_{[r]} \\
&= \left(\sum_{i=1}^{n} y_{[i]} - r \right) x_{[r]} \\
&= 0
\end{aligned}$$

$\phi(x, y)$ is bounded by 0, and equal to 0 for $x = (t, t, \ldots, t)$ since $\sum_{i=1}^{n} y_i = r$. The supremum is thus equal to 0.

We finally get the conjugate function:

$$f^*(y) = \begin{cases} 0 & \text{if } 0 \preceq y \preceq 1 \text{ and } \sum_{i=1}^{n} y_i = r \\ +\infty & \text{otherwise} \end{cases}$$

3. Piecewise-linear function on $\mathbb{R}$:

$$f(x) = \max_{i=1,\ldots,m} (a_i x + b_i),$$

with $a_1 \leq \cdots \leq a_m$ and no redundant function.

Answer: The conjugate of f is

$$f^*(y) = \begin{cases} (y - a_k) \left(\frac{b_k - b_{k+1}}{a_{k+1} - a_k} \right) - b_k & \text{if } y \in [a_k, a_{k+1}] \text{ for } k = 1, \ldots, m-1 \\ +\infty & \text{if } y \notin [a_1, a_m] \end{cases}$$

We start with determining the domain of the conjugate function.
Let $\phi(x, y) = yx - \max_{i=1,\ldots,m}(a_i x + b_i)$. Looking at the behavior of ϕ as $x \to +\infty$, we notice that the term in the max with the largest slope a_m will dominate. We get:

$$\phi(x, y) = yx - (a_m x + b_m) = (y - a_m)x - b_m$$

In order for $\phi(x, y)$ not to go to $+\infty$ as $x \to +\infty$, we must have $y - a_m \leq 0$, which implies $y \leq a_m$.
On the other hand, if $x \to -\infty$, then the term in the max with the smallest slope a_1 will dominate. $\phi(x, y)$ becomes:

$$\phi(x, y) = yx - (a_1 x + b_1) = (y - a_1)x - b_1$$

In order for $\phi(x, y)$ not to go to $+\infty$ as $x \to -\infty$, we must have $y - a_1 \geq 0$, which implies $y \geq a_1$.

In conclusion, the supremum is finite if y is in the closed interval $[a_1, a_m]$. Outside of this interval, $f^*(y) = +\infty$.

We can now calculate the supremum for $y \in [a_1, a_m]$.
Let's rewrite $\phi(x, y)$ as:

$$\begin{aligned}\phi(x, y) &= yx - \max_{i=1,\dots,m}(a_i x + b_i) \\ &= \min_{i=1,\dots,m}((y - a_i)x - b_i) \\ &= h(x)\end{aligned}$$

We notice that this function $h(x)$ is the minimum of m lines. $h(x)$ is thus a piecewise-linear concave function, as the minimum of linear functions. In order to find the $\sup_{x\in\mathbb{R}}$ of this function, we look at the slopes.
Let $c_i = y - a_i$. Since $y \in [a_1, a_m]$ and the a_i are ordered increasing, there must be some index k such that $a_k \leq y \leq a_{k+1}$.

- For $i \leq k$, we have $a_i \leq a_k \leq y$, which means that $c_i \geq 0$.
- For $i \geq k + 1$, we have $y \leq a_{k+1} \leq a_i$, which means that $c_i \leq 0$.

The sup of $h(x)$ must then occur as between k and $k+1$. We will find the breakpoint x_k where this happens:

$$(y - a_k)x_k - b_k = (y - a_{k+1})x_k - b_{k+1}$$

Solving for x_k, we get:

$$x_k = \frac{b_k - b_{k+1}}{a_{k+1} - a_k}$$

We calculate our supremum at this point and we get, for $y \in [a_k, a_{k+1}]$:

$$f^*(y) = (y - a_k)x_k - b_k = (y - a_k)\left(\frac{b_k - b_{k+1}}{a_{k+1} - a_k}\right) - b_k$$

The final conjugate function is:

$$f^*(y) = \begin{cases}(y - a_k)\left(\frac{b_k - b_{k+1}}{a_{k+1} - a_k}\right) - b_k & \text{if } y \in [a_k, a_{k+1}] \text{ for } k = 1, \dots, m-1 \\ +\infty & \text{if } y \notin [a_1, a_m]\end{cases}$$